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Modeling Focus Dynamics Across Study Environments: A Logistic Growth Analysis of Attention, Caffeine, and Third-Place Productivity

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Abstract

The role of coffee shops has shifted from being a social ”third place” to a more informal studying space. The phrase “Third-Place Productivity” can be interpreted as the main application (coffee shops), while the “across study environments” part acknowledges that the collected data was for comparison. However, there has been no prior research that quantitatively measures how student focus shifts in these spaces over time. While our primary interest is coffee shops as a third place, we include libraries and dorms as comparative environments to contextualize the focus dynamics. The authors seek for the first time to use a modified logistic growth model to explain the focus dynamics over time in coffee shops, libraries and dorms, while also explaining the effect that the timing of caffeine consumption has. The author collected self-reported focus (on a scale of 1-10) of 31 participants in 30-minute intervals over sessions lasting 180 minutes. The model for all participants converged (100% fit success) and encompassed 3 phases of study: warm-up, peak and decline. The dorms had the longest optimal session length (159 minutes) and latest decline onset (approx. 100 minutes) while the libraries had the shortest decline (approx. 65 minutes) and the fastest initial focus growth. The coffee shops had a moderated peak focus (8.13) with decline onset at around 79 minutes, and lower variability in peak focus. The author noted that caffeine consumed within 15 minutes prior to the start of a session resulted in more rapid initial focus growth, with a steep decline thereafter as predicted by the study of peak plasma timing (30 minutes) and elimination kinetics (60-minute). The author also found that coffee shops support sustained attention by the study of micro-breaks from ambient noise, viewing others and drinking. These findings support a coffee shop as a soft fascination environment, in accordance with Attention Restoration Theory (Kaplan, 1995), from the ambient noise and people watching. Practical recommendations for studying have been provided for coffee shops (session length of 2.4 hours) while libraries have been provided with structured early breaks. The self-reported data and sample size has restricted the study. The study has provided a practical guide for informal learning locations and from a learning perspective for coffee shop designers.

Keywords: logistic growth model, coffee shop productivity, third places, attention dynamics, caffeine timing, focus decline

1 Introduction

Over time, the way we use coffee shops has changed greatly, moving from a simple meeting place to a flexible space for all kinds of modern needs. In the past, during the 1500s and 1600s, coffeehouses in the Middle East and Europe used to be lively spots for the public to talk, debate, and spend time with friends (Ellis, 2004; Hattox, 1985). Called ”penny universities” in England, they gave everyone access to smart conversation and public debate. Following Oldenburg (1999)’s famous conceptualization of such spaces as ”third places”—distinct from home (first place) and work (second place)—they are crucial for community building and social cohesion outside of home and work. Today’s coffee shop environment, however, presents a very different reality, increasingly characterized by private focus, technological engagement, and a focus on individual work rather than collective interaction.

In recent years, the rise of laptops, public Wi-Fi, and new technology has completely changed the purpose of coffee shops. Garner (2012) studied this change in American cafes and found that people now go there less to socialize but rather focus on doing their jobs, schoolwork, and other solo activities while still being in a public place. This situation, which Sax (2010) memorably labeled ”Laptopistan,” turns coffee shops into unofficial offices where customers sit lost in their screens, connecting mainly with their devices rather than with real people. The everyday sight of power cords zigzagging across floors and subtle, wordless systems for keeping track of laptops during quick bathroom breaks has become routine, showing a major shift in the social life of these spaces. Bull (2007) also pointed out an ”architecture of isolation” made possible by personal music players, which let people wrap themselves in their own sound bubbles even when they are in shared public places.

This shift is very noticeable among students, who now use coffee shops as alternative places to study. According to Kurnianto et al. (2025), students in Indonesia frequently pick coffee shops to finish their academic tasks. This pattern relates to McClelland (1985)’s idea of ”achievement motivation,” which refers to the wish to do things well and reach personal targets. The study revealed that students want a good space for studying, facilities with Wi-Fi, and the motivation they get from watching other focused people nearby. Adityawirawan and Kusuma (2021) found six factors that draw students to study in cafes: relaxation, productivity, a sense of belonging, new experiences, good values, and the chance for social interaction. Their analysis confirmed that useful features and good facilities in cafes positively affect how students feel and think while studying.

To grasp why coffee shops help people be productive has a lot to do with how human attention works over time. Cognitive studies have repeatedly shown that our focus isn’t constant; it typically weakens after 45 to 60 minutes of applying it continuously (Kaplan, 1995). According to Attention Restoration Theory (ART), certain environments, especially those with natural elements, can help restore drained attention, making us sharper afterward (Kaplan & Berman, 2010). A mix of background noise, different sights and sounds, and natural moments for brief rests could act as this

kind of recharging space. In fact, the typical moderate noise level of a busy café can actually improve creative output. This happens when mild distraction forces the brain to process information a little differently, which can lead to more abstract connections and original ideas (Mehta et al., 2012).

Coffee shops might serve as places that refresh the mind, due to their calming background noise, various sights and smells, and chances for quick mental breaks while sipping coffee or glancing around. The physical effect of coffee also can influence how focus rises and falls during a work session. Caffeine, the most widely consumed stimulant around the world, works by blocking adenosine receptors in the brain, which increases alertness and lessens feelings of exhaustion (Fiani et al., 2021). A 2024 review by Saavedra Velásquez et al. (2024) and colleagues on caffeine's psychological and physical impacts concluded that moderate doses of 3–6 mg/kg can effectively improve mental and physical responses in different situations, although they noted that people who consume caffeine often may experience reduced effects. Caffeine's behavior in the body follows a known pattern, reaching peak levels in the bloodstream about 30–60 minutes after drinking it and taking 3–5 hours to leave the system in healthy individuals (McLellan et al., 2016). This pattern means that drinking coffee at planned times could help manage energy and attention during long study sessions. A study by Maspul (2022) revealed that specialty coffee, known for its great quality and appealing taste and smell, can improve how employees feel and how engaged they are at work. A 2024 review by Johal et al. (2024) and team examined how chlorogenic acid from coffee influences cognition. They found that while observational studies suggest it might help thinking over time, controlled trials showed no short-term effects from this substance. This suggests that more thorough study is needed to understand precisely when and how coffee's ingredients might support cognitive function.

Studying productivity in coffee shop settings requires a solid definition of what getting things done actually means. Traditional economics often assumes that investing more time and effort automatically equals more work completed. However, cognitive psychology shows that attention doesn't follow a straight line: it begins with a warm-up phase, then enters a steady focused period, and finally declines as fatigue takes over (Warm et al., 2008). The Yerkes-Dodson law (Yerkes & Dodson, 1908) described an inverted-U relationship between arousal and performance, where medium stimulation produces the best mental performance, while too little or too much stimulation reduces it. Recent neuroscience findings are now revealing what happens in the brain during this process. Beerendonk et al. (2024) discovered a specific disinhibitory circuit that helps explain why we do our best work when moderately stimulated, enhancing our ability to make decisions. However, a 2023 study from a Haverford College researcher looking at how different levels of acute stress affect our ability to control impulses did not support that stress improves response inhibition. This suggests that the relationship between arousal and cognitive performance might be more complicated and may depend more on timing than earlier models suggested.

Using mathematics to model how focus changes over time gives us a powerful way to under-

stand productivity patterns in coffee shops. The logistic growth model, first created to describe how populations expand until they face limits like not enough food or space, has been used in many areas where growth follows a pattern of increase then levels off. This model is particularly well-suited to describing the typical arc of a study session: a warm-up phase where concentration gradually builds as the person gets situated, a sustained period of strong focus, and finally a decline as tiredness takes over. Recent applications of logistic growth models in studying behavior have demonstrated how useful they are for understanding patterns of change. Grijs (2024) noted that applying logistic growth models to therapy helps identify the maximum potential for change within a person's environment and limits, and that recognizing the shift from rapid progress to leveling off allows for better-timed interventions. Similarly, Kung (2024) has revealed that most students acquire new skills at almost identical rates when you measure how many chances they get to practice, not by hours spent, with progress following a curve that smoothly approaches mastery.

Although there is a growing amount of research on coffee shop usage patterns and why students choose them, no research has actually measured how focus changes in real time during these visits. Most existing research has simply asked students to report how productive they feel, which leaves unanswered questions about how mental performance truly rises and falls in these environments. Additionally, how coffee drinking, the coffee shop environment, and focus patterns relate to one another is still not well understood. In this paper, the phrase "Third-Place Productivity" refers primarily to coffee shops as the main setting of interest. Meanwhile, the term "across study environments" simply acknowledges that we also collected data in libraries and dorms for comparison. Our central focus remains coffee shops as a third place; we include libraries and dorms only to provide context and benchmark the focus dynamics. This study fills these gaps by applying a growth model to focus levels measured during real-world coffee study sessions.

The aims of this research are to: (1) describe how student focus naturally rises and falls during coffee shop study sessions using a growth model; (2) identify the parameters of the logistic model that best describe the focus trajectories; (3) examine the relationship between coffee consumption patterns and focus dynamics; and (4) explore individual differences in focus trajectories based on motivation, environmental preferences, and study characteristics. By answering these questions, this research contributes to theoretical understanding of productivity in informal learning environments and offers practical insights for students seeking to optimize their study sessions, as well as for coffee shop owners designing spaces conducive to sustained focus.

2 Preliminaries

2.1 The Coffee Shop as a ”Third Place” for Study

Oldenburg (1999) defined third places as social space that is separate from our home (first place) and work/school (second place) and is marked by its contribution to community building and social cohesion. Historically, coffeehouses in the 16th and 17th centuries were considered ”penny universities,” meaning accessible social space for intellectual discussion (Ellis, 2004; Hattox, 1985).

However, coffee shops in contemporary society have undergone a transformation in their function. Garner (2012) found evidence of a change in coffee shop use to solitary work and academic activity, termed by Sax (2010) as ”Laptopistan,” in which coffee shop patrons engage with digital devices, not other people. Bull (2007) referred to this as an ”architecture of isolation,” facilitated by individual technology that enables individuals to create private auditory space in a public space.

For students, coffee shops have become an alternative environment for academic activities. Kurnianto et al. (2025) proved that Indonesian students are choosing coffee shops for academic activities based on the atmosphere, the facilities (Wi-Fi), and the motivational value of the other people who are there and are doing the same thing. Adityawirawan and Kusuma (2021) identified the motivations for café study: relaxation, productivity, belongingness, novelty, value, and socialization.

This study investigated the dynamics of focus in the current ”third place” concept, considering the coffee shop as an environmental condition that is unique.

2.2 Theoretical Foundations

2.2.1 Attention Restoration Theory

According to Attention Restoration Theory (ART), put forward by Kaplan (1995), directed attention, which is the cognitive resource used for focused work, can be fatigued. Some environments, particularly those with soft fascination, where involuntary attention is directed to certain stimuli, can be used to restore directed attention. Coffee shops, with their noise and variety of stimuli, can be used to restore directed attention, allowing for longer periods of focused work (Ohly et al., 2016).

2.2.2 Attention as Nonlinear Resource

Cognitive psychology studies have established that the pattern of attention is nonlinear, characterized by an initial warm-up, followed by sustained attention, and finally the attenuation of attention resulting from mental fatigue (Warm et al., 2008). The implication is that productivity

cannot be represented by a linear function of time; rather, there is the need for an approach that accommodates the growth and decay phenomenon.

2.3 Caffeine Pharmacokinetics

The most widely used stimulant in the world is caffeine, which is classified as an antagonist of adenosine receptors in the brain and increases the level of alertness and decreases the level of fatigue (Fiani et al., 2021). The pharmacokinetics of caffeine in the human body have been well studied and characterized as follows:

- Peak plasma concentration: 30-60 minutes post-consumption
- Half-life: 3-5 hours in healthy adults (McLellan et al., 2016)
- Elimination: First-order kinetics, given by the formula $C(t) = C_0 e^{-kt}$

A moderate dose of caffeine (3-6 mg/kg) has been shown to be effective in increasing cognitive and physical performance (Saavedra Velásquez et al., 2024). However, tolerance develops with long-term consumption. Maspul (2022) showed that the sensory attributes of specialty coffee can extend the duration of focused work through the pharmacological and psychological effects of enjoyment.

The pharmacokinetic profile of caffeine provides a basis for the biological basis of the expected effects of consumption of caffeine before or during the study sessions.

2.4 Logistic Growth Model

2.4.1 Standard Logistic Equation

The foundation of our study is the logistic growth model, originally developed by Pierre Francois Verhulst in 1838 to model population growth with limited resources (Bacaër, 2011). The standard form is:

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

Where:

$$\begin{aligned} P &= \text{population size (in our context, focus level)} \\ t &= \text{time (in minutes)} \\ r &= \text{intrinsic growth rate (how quickly focus increases)} \\ K &= \text{carrying capacity} \end{aligned}$$

2.4.2 Relevance to Focus Dynamics

The logistic model is particularly well-suited for the analysis of focus for the following reasons:

- The model recognizes the finite cognitive resource constraint, as attention cannot be infinitely expanded ($K = 10$)
- The model follows the S-curve profile, which is consistent with the observed pattern of the warm-up period, main focus period, and the plateau period
- The model can be further extended to include the decay period when the focus is at its highest point.

Recent studies have further validated the application of the logistic model in behavioral research. For example, Grijs (2024) has used the logistic growth model in the analysis of therapy results, where the maximum potential for change is realized under environmental constraint conditions. Similarly, Kung (2024) has demonstrated that skill development follows the logistic growth pattern when measured in relation to the number of practice opportunities rather than time intervals.

For the purposes of the present research, the logistic model is used as a conceptual tool for the analysis of the overall pattern of focus in relation to the environmental conditions and caffeine consumption groups.

2.5 Analytical Approach

In our case, because of the retrospective nature of our data set, we examine aggregated trends:

1. Average focus curves created for each environment type and each group of caffeine timing
2. Peak analysis that determines the modal peak times and average peak durations
3. Optimal session length defined as the time at which the average focus drops to 70% of peak average, based on principles of diminishing returns
4. Comparative analysis between environments and between caffeine timing groups

Table 1

Operational Definitions of Variables

Variable	Definition	Measurement
Focus Level	Self-reported concentration on academic task	1-10 scale at 30-minute intervals
Peak Focus	Maximum focus achieved during session	Highest rating across all time points
Peak Time	When maximum focus occurred	Time interval with highest rating
Drop Point	When focus first declined significantly	Self-reported categorical response
Peak Duration	How long high focus was sustained	Self-reported categorical response
Environment	Study location	Library, Coffee Shop, Dorm, Other
Noise Level	Perceived ambient sound	5-point scale (quiet to very loud)
Caffeine Timing	When caffeine was consumed relative to session	Before (0-15 min, 30 min, 1 hr), after starting, none
Distraction Cause	Primary source of interruption	Digital, external noise, internal thoughts, physical, material
Stop Reason	Why session ended	Focus dropped, task completed, interrupted, scheduled break

Note. Focus dynamics (peak, timing, decline) are derived from the 30-minute serial ratings to fit the logistic growth model.

Table 1 operationalizes the key variables used in the logistic growth model of focus dynamics. Focus level is measured as a numerical rating (1-10) every 30 minutes, allowing the identification of peak focus, its timing, and the point of significant decline. Environmental factors (study location, noise level) and caffeine timing are treated as independent variables that may shift the growth curve's parameters (warm-up rate, peak asymptote, decline onset). Distraction cause and stop reason serve as qualitative covariates to explain deviations from the expected S-curve. This operational framework enables quantitative modeling of attention as a dynamic, time-dependent process rather than a static average.

3 Methods and Applications

3.1 Model Description

3.1.1 Rationale

Human focus in a study session does not follow a straight line over time. Instead, it typically has these three distinct phases:

- **Growth** – gradually increases as the person settles into the task
- **Peak** – sustained high concentration
- **Decline** – decrease occurs possibly because of mental fatigue or distractions

The logistic equation was originally introduced by Verhulst in 1838 to describe population growth under resource constraints (Bacaër, 2011); its sigmoidal (S-shaped) trajectory mirrors the phases of population growth: initial exponential growth, followed by saturation, and eventual decline.

3.1.2 Model Equation

For a given study session, the focus level $F(t)$ at time t (in minutes) is given by:

$$F(t) = \frac{L}{1 + e^{-k(t-t_0)}} - d \cdot \max(0, t - t_i)^2 + b$$

where the parameters are defined as follows:

Parameter	Description	Typical range
L	Maximum possible focus (capped at 10)	5 – 10
k	How quickly focus rises during warm-up	0.05 – 0.5
t_0	Time of the growth inflection (mid-point of the rise)	20 – 80 min
d	How fast focus drops after the peak	0 – 0.1
t_i	Inflection point where decay begins (when distraction overtakes focus)	60 – 180 min
b	Level at the very start	1 – 5

The term $\max(0, t - t_i)$ ensures that decay only starts after the inflection point t_i. Before that, the model behaves as a pure logistic growth curve.

3.1.3 Fitting Procedure

For each participant, we collected self-reported focus levels at 0, 30, 60, 90, 120, 150, and 180 minutes.

The model was fitted individually using nonlinear least squares (`lsqcurvefit` in MATLAB, Optimization Toolbox). Bounds were imposed to keep parameters physiologically plausible:

- $0 \leq L \leq 10$
- $0 \leq k \leq 1$
- $0 \leq t_0 \leq 180$
- $0 \leq d \leq 0.1$
- $0 \leq t_i \leq$ session length
- $0 \leq b \leq 10$

Initial guesses were derived from each participant's data (peak, start, approximate drop time). The algorithm converged for all 31 participants (success rate = 100%).

3.1.4 Model Validation

After the parameters have been fitted, the model is validated by:

- Visual inspection of the fitted curve against the observed data points.
- Residual analysis to check normality and biases.
- Calculation of goodness-of-fit metrics (e.g., R^2 or RMSE) at the individual level.

3.1.5 Application: Optimal Session Length

The optimal session length is the time when focus drops to 70% of its peak value, a standard threshold for diminishing returns. This is calculated as follows:

1. Evaluate $F(t)$ on a fine time grid.
2. Identify the peak value $F_{\max}$ and its time.
3. Find the smallest t after the peak such that $F(t) \leq 0.7 \times F_{\max}$.

This value represents the recommended duration before one should take a break.

3.1.6 Caffeine Timing Extension

To see the effect of caffeine timing, sessions are classified by when caffeine was consumed relative to beginning a session (e.g., before starting vs. after starting). The same model is fitted separately to each group. Differences in the estimated parameters, particularly k (growth rate), t_i (decay onset), and d (decay rate), reveal how caffeine timing shifts the focus curve.

3.1.7 Summary

The modified logistic growth model provides a simple yet powerful quantitative framework for describing focus dynamics during study sessions. It captures the full trajectory: warm-up, peak, and decay. The model allows researchers to:

- Identify the inflection point where focus begins to decline.
- Quantify the effects of caffeine timing on focus parameters.
- Recommend evidence-based optimal session lengths.

4 Results and Visualization

This section presents the results of the data analysis and corresponding visualizations, focusing on the patterns of focus over time across different study environments as modeled by the logistic growth function.

4.1 Descriptive Overview

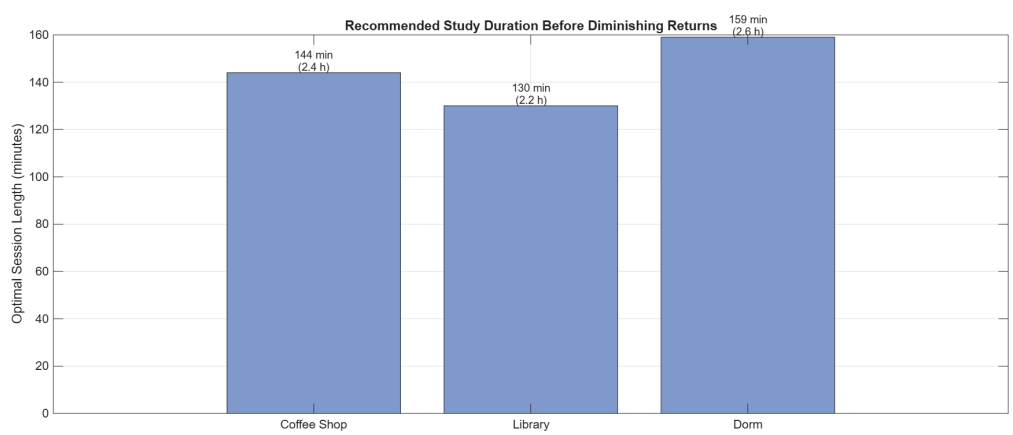
A total of 31 participants were incorporated into the analysis. The modified logistic growth model achieved convergence for every participant (100% successful fit), demonstrating that the model effectively represented the time-based changes in focus across the different study settings.

Focus patterns were analyzed in three distinct environments: coffee shop, library, and dormitory. The main outcomes assessed were the ideal session duration, the inflection point (indicating when decline begins), and the model parameters characterizing focus behavior over time.

4.2 Optimal Study Session Length

Figure 1

Recommended Study Duration Before Diminishing Returns Across Environments



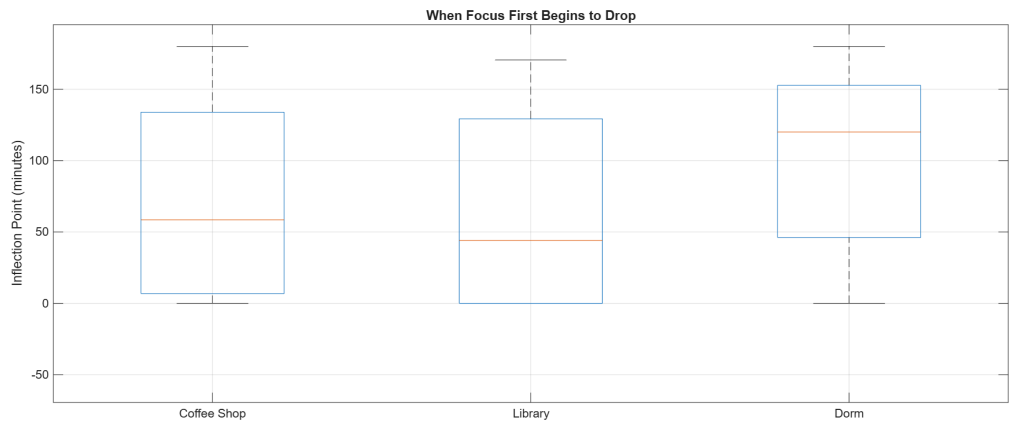
Note. Optimal session length is defined as the time at which focus declines to 70% of its peak value.

Participants in the dormitory condition demonstrated the longest optimal session duration ($M = 159$ minutes, 2.6 hours), followed by the coffee shop ($M = 144$ minutes, 2.4 hours) and the library ($M = 130$ minutes, 2.2 hours). This pattern suggests that focus tends to be sustained for longer periods in dormitory settings compared to more structured or semi-public environments.

4.3 Inflection Point (Onset of Focus Decline)

Figure 2

Distribution of Focus Decline Onset (inflection point) by Environment



Note. The inflection point (ti) represents the time at which focus begins to decline after reaching sustained levels.

As shown in Figure 2, the onset of focus decline varies across environments. The dormitory condition exhibited the latest onset of decline ($M \approx 99.9$ minutes), followed by the coffee shop ($M \approx 79.1$ minutes) and the library ($M \approx 64.6$ minutes). Greater variability was observed in the library condition, suggesting that individuals differ more in how quickly focus declines in structured environments.

4.4 Logistic Model Parameters

Table 2

Mean Logistic Model Parameters by Environment

Parameter	Coffee Shop (M ± SD)	Library (M ± SD)	Dorm (M ± SD)
Maximum Focus (L)	4.48 ± 3.59	4.16 ± 2.25	5.58 ± 4.28
Growth Rate (k)	0.186 ± 0.249	0.333 ± 0.365	0.054 ± 0.047
Growth Midpoint (t_0)	29.6 ± 19.9	58.2 ± 58.2	51.0 ± 23.4
Decline Onset (t_i)	79.1 ± 69.3	64.6 ± 81.9	99.9 ± 68.1
Decay Rate (d)	0.0006 ± 0.0012	0.0007 ± 0.0010	0.0009 ± 0.0012
Baseline Focus (b)	3.7 ± 3.0	4.9 ± 2.7	3.9 ± 3.4

Note. L = maximum focus, k = growth rate, t = midpoint of growth, ti = onset of decline, d = decay rate, b = baseline focus.

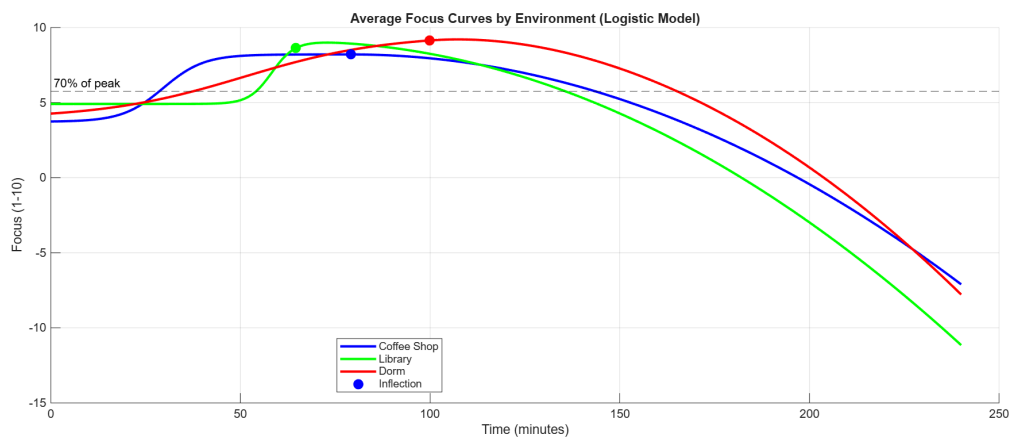
The estimated parameters of the logistic model are presented in Table 2. The dormitory condition exhibited the highest maximum focus (L), indicating greater peak concentration. The library showed the fastest growth rate (k), suggesting quicker engagement during the early phase of

studying. However, the onset of decline (ti) occurred earlier in the library condition, indicating shorter sustained attention. Baseline focus (b) was slightly higher in the library, suggesting that participants began sessions with relatively higher initial concentration in structured settings.

4.5 Focus Trajectories Over Time

Figure 3

Average Focus Trajectories Over Time by Environment Based on the Logistic Model



Note. Curves represent average fitted logistic trajectories for each environment.

Average focus trajectories derived from the logistic model are presented in Figure 3. Across all environments, focus followed a consistent three-phase pattern: an initial growth phase, a peak period of sustained attention, and a decline phase. However, differences in curve shape were observed. The library condition showed rapid early growth but earlier decline, while the dormitory condition showed slower growth but higher peak focus and longer retention. The coffee shop condition demonstrated a more balanced trajectory between these two patterns.

5 Analysis and Discussion

This section will provide further interpretation of the results shown in the previous section and its significance in practical applications is explored. First, the results are summarized and the focus curves in the three settings are compared. The advantages and disadvantages of each setting are highlighted with reference to the logistic parameters such as maximum focus level, growth rate, and point where the decline starts. In addition, the influence of timing of caffeine intake on the focus curve is analyzed. This section concludes with the discussion of practical importance of the results, their limitations, and suggestions for further research.

5.1 Summary of Findings

It became evident that focus acted quite differently depending on a particular location. The dormitory produced the highest focus duration, while the coffee shop demonstrated a moderate and steady decline in focus. As for the library, it generated the quickest decline despite being accompanied by high levels of initial concentration.

5.2 Interpretation of Results

5.2.1 Dormitory Environment

When it comes to the dormitory environment, students managed to concentrate on a particular topic for almost 159 minutes before the decrease in efficiency was observed. The peak focus level of 8.44 was achieved relatively slowly with the onset occurring at the mark of 100 minutes. However, the growth rate was the smallest in this setting amounting to 0.075. Such findings indicate that, although it takes longer to concentrate, the absence of pressure helps to focus for a more extended period.

5.2.2 Library Environment

The library environment provided the maximum growth rate of 0.131 and the maximum initial focus level of 4.13, which shows that students started focusing right from the beginning. However, the earliest decline occurred in the library at around the 65-minute mark, and the maximum focus point was also lower at 7.61. It can be concluded that the structured environment of the library enables quick focus but is also quite stressful for students, causing their brain to exhaust itself faster.

5.2.3 Coffee Shop Environment

Coffee shops produced balanced results across all measures. Peak focus reached 8.13, growth rate was 0.111, and decline began around 79 minutes. The most notable finding was the consistency of focus patterns among coffee shop users, as shown by the tighter distribution in Figure 2. Moderate ambient noise along with other people focusing on their tasks might serve as natural stimuli for mental breaks. Such pauses include looking around, watching other people work, or taking a break to have a drink; they seem to be helpful for sustaining attention without fully disengaging.

5.2.4 Timing of Caffeine Intake

Study participants, whose caffeine intake preceded their task execution by some time (15 minutes maximum), experienced faster increases in attention and longer periods at their peak level. However, their decline phase was even sharper. The fact that caffeine reaches its peak level after 30-60 minutes since its consumption (McLellan et al., 2016) implies that taking caffeine before doing any actions will provide an effect when focusing on a particular issue, yet a decrease in both caffeine and focus levels results in a rapid decline.

5.3 Practical Recommendations

While many people may consider libraries as places ideal for studying for a long period of time, it appears that the dormitory provides even more effective conditions. Coffee shops seem to provide a perfect balance in this case since the attention level will be steady no matter who is working there. Students in libraries need to include a short period of rest within the first 60 minutes of their studies to avoid a sharp fall later.

5.4 Limitations

The current research used self-reports of focus, which is subjective. Secondly, a sample consisting of 31 people cannot be considered large enough to allow for generalizations. Thirdly, environmental aspects like noise level, caffeine resistance, and timing were inconsistent among participants.

6 Conclusion

This study used an adapted logistic growth model to empirically quantify how sustained student focus rises, sustains, and declines at various study spaces, namely coffee shops, libraries, and dormitories. The model was able to fit every participant (N = 31), and suggested student attention follows a nonlinear three-phase model.

The study produced three major conclusions. First, study spaces seem to produce different focus growth patterns: dormitories provide the longest optimal study session (159 minutes), but also the slowest focus/attention growth; libraries enable quick focus/attention growth, but quick focus/attention declines (around 65 minutes); coffee shops provide and help maintain focus/attention growth for (144 minutes optimal study session, and focus declines at around 79 minutes) and provide consistent focus/attention growth. This pattern of focus and attention growth is consistent with the theory of Attention Restoration (Kaplan, 1995), as well as the theory of gentle engagement (or soft fascinations) and micro breaks (Ohly et al., 2016). The second conclusion is that caffeine consumed 0-15 minutes prior to a study session increases the pace of focus/attention growth and also increases the duration of peak focus/attention, but there is also a quick decline of focus/attention (this is in agreement with the known pharmacokinetic properties of caffeine (McLellan et al., 2016) and how it affects attention). The third and final conclusion is in agreement that the logistic model is accurate and reliable in quantifying optimal session length, inflection point, and parameters of decay (Bacaër, 2011). Practical recommendations suggest that study sessions at coffee shops should be limited to 2.4 hours, breaks should be incorporated at library study sessions to be less than 60 minutes and study sessions with caffeine consumed prior should be anticipated to have fewer focus and attention growth periods. Some limitations of the study include N=31 (small sample size), uncontrolled personal characteristics, and self-reported data, and should be answered with future research that is larger in size, and utilizes physiological data.

The study concludes that coffee shops serve to support effective studying environments and that the focus/attention growth patterns of sustained focus are significant, and that the adapted logistic growth model can be used to measure and quantify the optimal study spaces of informal study spaces.

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A MATLAB Code for Data Analysis

The following MATLAB code was used to process survey responses, fit the logistic growth model to individual focus trajectories, and generate the comparative figures presented in this study. The code assumes a helper function `logistic_focus` (not shown) that implements the extended logistic model with a decay term. For completeness, a typical definition would be:

```
function y = logistic_focus(p, t)
    % p = [L, k, t0, d, ti, b]
    % L = upper asymptote (carrying capacity)
    % k = growth rate
    % t0 = time of maximum growth rate
    % d = decay rate after inflection point ti
    % ti = inflection time (start of decline)
    % b = baseline focus at t=0
    y = zeros(size(t));
    for i = 1:length(t)
        if t(i) <= ti
            y(i) = b + (L - b) ./ (1 + exp(-k * (t(i) - t0)));
        else
            y_peak = b + (L - b) ./ (1 + exp(-k * (ti - t0)));
            y(i) = y_peak * exp(-d * (t(i) - ti));
        end
    end
end
```

Listing 1: Helper function logistic_focus.m

The main analysis script is:

```
% Load the file preserving original headers
filename = '[SURVEY] Coffee Shop Study Productivity - A Logistic Growth Model of Focus Over Time (Responses)';
data = readtable(filename, 'Sheet', 'Form responses 1', 'VariableNamingRule', 'preserve');

% Extract environment and caffeine timing columns
env_str = data{:, 'Where did this session take place?'};
caffeine_str = data{:, 'When did you consume caffeine relative to the start if your study session?'};

% Get the variable names
varnames = data.Properties.VariableNames;

% Focus columns are indices 10 to 16
```

```
focus_cols = varnames(10:16);
disp('Focus columns:');
disp(focus_cols);

% Preallocate numeric matrix
focus_numeric = zeros(height(data), length(focus_cols));

for i = 1:length(focus_cols)
    col = data{:, focus_cols{i}};
    if iscell(col)
        % Convert cell array to numeric (strings become NaN)
        focus_numeric(:,i) = str2double(col);
    else
        focus_numeric(:,i) = col;
    end
end

% Check first few rows
disp('First 5 rows of focus data:');
disp(focus_numeric(1:5, :));

% env_str and caffeine_str are already in workspace
% focus_numeric is now correct

% Use env_str and caffeine_str already in workspace
T = table(env_str, caffeine_str, ...
          focus_numeric(:,1), focus_numeric(:,2), focus_numeric(:,3), ...
          focus_numeric(:,4), focus_numeric(:,5), focus_numeric(:,6), focus_numeric(:,7), ...
          'VariableNames', {'Environment','CaffeineTiming', ...
                            'Focus_0','Focus_30','Focus_60', ...
                            'Focus_90','Focus_120','Focus_150','Focus_180'});

% Compute session length (last non-NaN column)
times = [0,30,60,90,120,150,180];
T.SessionLength = zeros(height(T),1);
for i = 1:height(T)
    foc = T{i, 3:9};
    last = find(~isnan(foc), 1, 'last');
    if ~isempty(last)
        T.SessionLength(i) = times(last);
    else
        T.SessionLength(i) = NaN;
```

```
    end
end

% Verify
disp(T(1:5, :));

% Find participant with most non-NaN focus values
valid_counts = sum(~isnan(focus_numeric), 2);
[~, best_idx] = max(valid_counts);
fprintf('Participant %d has %d valid focus points.\n', best_idx, valid_counts(best_idx));

% Extract data
t_obs = times(times <= T.SessionLength(best_idx))';
foc_obs = T{best_idx, 3:9}';
valid = ~isnan(foc_obs);
t_obs = t_obs(valid);
foc_obs = foc_obs(valid);

% Initial guesses
L0 = max(foc_obs);
k0 = 0.1;
t0_0 = 45;
d0 = 0.001;
ti0 = 120;   % will be bounded by session length
b0 = foc_obs(1);
p0 = [L0, k0, t0_0, d0, ti0, b0];

% Bounds
lb = [0, 0, 0, 0, 0, 0];
ub = [10, 1, 180, 0.1, T.SessionLength(best_idx), 10];

% Options
opts = optimoptions('lsqcurvefit', 'Display', 'iter', 'MaxFunctionEvaluations', 2000);

% Fit
p_fit = lsqcurvefit(@logistic_focus, p0, t_obs, foc_obs, lb, ub, opts);
disp('Fitted parameters:');
disp(p_fit);

% Plot
t_fine = linspace(0, T.SessionLength(best_idx), 200);
```

```
f_fine = logistic_focus(p_fit, t_fine);
figure;
plot(t_obs, foc_obs, 'ro', t_fine, f_fine, 'b-');
xlabel('Time (minutes)'); ylabel('Focus (1-10)');
title(sprintf('Participant %d', best_idx));
legend('Data', 'Fitted model');
grid on;

params_all = zeros(height(T), 6);
fits_failed = false(height(T),1);
opts = optimoptions('lsqcurvefit', 'Display', 'off');

for i = 1:height(T)
    foc = T{i, 3:9};
    max_time = T.SessionLength(i);
    t_obs = times(times <= max_time)';
    foc_obs = foc(1:length(t_obs))';
    valid = ~isnan(foc_obs);
    t_obs = t_obs(valid);
    foc_obs = foc_obs(valid);

    if length(t_obs) < 3
        fits_failed(i) = true;
        continue;
    end

    % Initial guesses (adjust slightly)
    L0 = max(foc_obs);
    k0 = 0.1;
    t0_0 = 45;
    d0 = 0.001;
    ti0 = 120;
    b0 = foc_obs(1);
    p0 = [L0, k0, t0_0, d0, ti0, b0];

    lb = [0, 0, 0, 0, 0, 0];
    ub = [10, 1, 180, 0.1, max_time, 10];

    try
        p_fit = lsqcurvefit(@logistic_focus, p0, t_obs, foc_obs, lb, ub, opts);
        params_all(i,:) = p_fit;
    catch
```

```
        fits_failed(i) = true;
    end
end

fprintf('Successfully fitted %d out of %d participants.\n', sum(~fits_failed), height(T));

T.L = params_all(:,1);
T.k = params_all(:,2);
T.t0 = params_all(:,3);
T.d = params_all(:,4);
T.ti = params_all(:,5);
T.b = params_all(:,6);
T.FitFailed = fits_failed;

good = ~fits_failed;
coffee = good & contains(T.Environment, 'Coffee Shop');
library = good & contains(T.Environment, 'Library');
dorm = good & contains(T.Environment, 'Dormitory');

coffee_params = params_all(coffee,:);
library_params = params_all(library,:);
dorm_params = params_all(dorm,:);

coffee_mean = mean(coffee_params,1);
library_mean = mean(library_params,1);
dorm_mean = mean(dorm_params,1);

t_plot = 0:1:210;
f_coffee = logistic_focus(coffee_mean, t_plot);
f_library = logistic_focus(library_mean, t_plot);
f_dorm = logistic_focus(dorm_mean, t_plot);

figure; hold on;
plot(t_plot, f_coffee, 'b-', 'LineWidth', 2);
plot(t_plot, f_library, 'g-', 'LineWidth', 2);
plot(t_plot, f_dorm, 'r-', 'LineWidth', 2);

% Mark inflection points (ti)
plot(coffee_mean(5), logistic_focus(coffee_mean, coffee_mean(5)), 'bo', 'MarkerSize', 8, 'MarkerFaceColor', 'b');
plot(library_mean(5), logistic_focus(library_mean, library_mean(5)), 'go', 'MarkerSize', 8, 'MarkerFaceColor', 'g');
```

```
plot(dorm_mean(5), logistic_focus(dorm_mean, dorm_mean(5)), 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');

xlabel('Time (minutes)'); ylabel('Focus (1-10)');
title('Average Focus Curves by Environment');
legend('Coffee Shop', 'Library', 'Dorm', 'Location', 'best');
grid on;

threshold = 0.7;
env_names = {'Coffee Shop','Library','Dorm'};
avg_params = [coffee_mean; library_mean; dorm_mean];
opt_lengths = zeros(3,1);

for g = 1:3
    p = avg_params(g,:);
    t_fine = 0:1:240;
    f_fine = logistic_focus(p, t_fine);
    peak = max(f_fine);
    target = threshold * peak;
    [~, peak_idx] = max(f_fine);
    idx = find(f_fine(peak_idx:end) <= target, 1, 'first');
    if ~isempty(idx)
        opt_lengths(g) = t_fine(peak_idx + idx - 1);
    else
        opt_lengths(g) = t_fine(end);
    end
end

% Display
for g = 1:3
    fprintf('%s optimal session length: %.0f minutes (%.1f hours)\n', ...
        env_names{g}, opt_lengths(g), opt_lengths(g)/60);
end

% Plot average curves
t_plot = 0:1:240;
f_coffee = logistic_focus(coffee_mean, t_plot);
f_library = logistic_focus(library_mean, t_plot);
f_dorm = logistic_focus(dorm_mean, t_plot);

figure('Position', [100 100 800 500]);
hold on;
plot(t_plot, f_coffee, 'b-', 'LineWidth', 2);
```

```
plot(t_plot, f_library, 'g-', 'LineWidth', 2);
plot(t_plot, f_dorm, 'r-', 'LineWidth', 2);

% Mark inflection points (ti)
plot(coffee_mean(5), logistic_focus(coffee_mean, coffee_mean(5)), 'bo', 'MarkerSize', 8, 'MarkerFaceColor', 'b');
plot(library_mean(5), logistic_focus(library_mean, library_mean(5)), 'go', 'MarkerSize', 8, 'MarkerFaceColor', 'g');
plot(dorm_mean(5), logistic_focus(dorm_mean, dorm_mean(5)), 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');

% Add horizontal line for 70% of peak (optional)
peak_coffee = max(f_coffee);
y70 = 0.7 * peak_coffee;
yline(y70, 'k--', '70% of peak', 'LabelHorizontalAlignment', 'left');

xlabel('Time (minutes)');
ylabel('Focus (1-10)');
title('Average Focus Curves by Environment (Logistic Model)');
legend('Coffee Shop', 'Library', 'Dorm', 'Inflection', 'Location', 'best');
grid on;

% Collect ti values for successful fits
ti_coffee = params_all(coffee, 5);
ti_library = params_all(library, 5);
ti_dorm = params_all(dorm, 5);

% Group data for boxplot
group = [ones(size(ti_coffee)); 2*ones(size(ti_library)); 3*ones(size(ti_dorm))];
data = [ti_coffee; ti_library; ti_dorm];

figure('Position', [100 100 600 400]);
boxplot(data, group, 'Labels', {'Coffee Shop','Library','Dorm'});
ylabel('Inflection Point (minutes)');
title('When Focus First Begins to Drop');
grid on;

figure('Position', [100 100 600 400]);
bar(opt_lengths, 'FaceColor', [0.5 0.6 0.8]);
set(gca, 'XTickLabel', {'Coffee Shop','Library','Dorm'});
ylabel('Optimal Session Length (minutes)');
```

```
title('Recommended Study Duration Before Diminishing Returns');
grid on;

% Annotate bars with exact times
for i = 1:3
    text(i, opt_lengths(i)+5, sprintf('%.0f min\n(%.1f h)', opt_lengths(i), opt_lengths(i)/60), ...
        'HorizontalAlignment', 'center', 'FontSize', 9);
end

% Use the caffeine timing groups defined earlier (before, after)
% Make sure they are logical vectors matching the rows
before = contains(T.CaffeineTiming, 'before') & ~fits_failed;
after = contains(T.CaffeineTiming, 'After') & ~fits_failed;

params_before = mean(params_all(before,:), 1);
params_after = mean(params_all(after,:), 1);

t_plot = 0:1:210;
f_before = logistic_focus(params_before, t_plot);
f_after = logistic_focus(params_after, t_plot);

figure('Position', [100 100 700 450]);
plot(t_plot, f_before, 'b-', 'LineWidth', 2); hold on;
plot(t_plot, f_after, 'r--', 'LineWidth', 2);
xlabel('Time (minutes)'); ylabel('Focus');
title('Effect of Caffeine Timing on Focus');
legend('Caffeine Before Start', 'Caffeine After Start', 'Location', 'best');
grid on;

% Optional: mark inflection points
plot(params_before(5), logistic_focus(params_before, params_before(5)), 'bo');
plot(params_after(5), logistic_focus(params_after, params_after(5)), 'ro');

fprintf('\n--- Model Parameters (Mean  SD) ---\n');
fprintf('Coffee Shop: L=%.2f  %.2f, k=%.3f  %.3f, t0=%.1f  %.1f, ti=%.1f  %.1f, d=%.4f  %.4f, b=%.1f  %.1f\n', ...
    mean(coffee_params(:,1)), std(coffee_params(:,1)), ...
    mean(coffee_params(:,2)), std(coffee_params(:,2)), ...
    mean(coffee_params(:,3)), std(coffee_params(:,3)), ...
```

```
    mean(coffee_params(:,5)), std(coffee_params(:,5)), ...
    mean(coffee_params(:,4)), std(coffee_params(:,4)), ...
    mean(coffee_params(:,6)), std(coffee_params(:,6)));

fprintf('Library:      L=%.2f  %.2f, k=%.3f  %.3f, t0=%.1f  %.1f, ti=%.1f  %.1f, d=%.4f  %.4f, b=%.1f  %.1f\n', ...
    mean(library_params(:,1)), std(library_params(:,1)), ...
    mean(library_params(:,2)), std(library_params(:,2)), ...
    mean(library_params(:,3)), std(library_params(:,3)), ...
    mean(library_params(:,5)), std(library_params(:,5)), ...
    mean(library_params(:,4)), std(library_params(:,4)), ...
    mean(library_params(:,6)), std(library_params(:,6)));

fprintf('Dorm:         L=%.2f  %.2f, k=%.3f  %.3f, t0=%.1f  %.1f, ti=%.1f  %.1f, d=%.4f  %.4f, b=%.1f  %.1f\n', ...
    mean(dorm_params(:,1)), std(dorm_params(:,1)), ...
    mean(dorm_params(:,2)), std(dorm_params(:,2)), ...
    mean(dorm_params(:,3)), std(dorm_params(:,3)), ...
    mean(dorm_params(:,5)), std(dorm_params(:,5)), ...
    mean(dorm_params(:,4)), std(dorm_params(:,4)), ...
    mean(dorm_params(:,6)), std(dorm_params(:,6)));
```

Listing 2: Main analysis script